Rahul Kumar Kundu

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Personal Website | **Google Scholar** | **ORCID** | **GitHub** | **LinkedIn**

Research Interests

- **Trustworthy AI:** Robustness, Generalization & Reliability.
- **AI Security:** AI-based Cybersecurity, Foundation Model Security & Privacy.
- **Multi-Objective Optimization:** Balancing Model Performance, Fairness & Robustness.

Education

Bachelor of Science (B.Sc.) in Electrical & Computer Engineering (ECE) **30 June 2025**
Rajshahi University of Engineering & Technology (RUET) Rajshahi, Bangladesh

- CGPA **3.79/4.00** and graduated with First Class Honors, ranking **7th** in the department out of 61 students.
- ***Undergraduate Thesis:*** ED-Net, Colorectal Polyp Segmentation *Published in BSPC, 2026*

Journal Publications

[1] **Kundu RK**, Kibria HB, Ahamed MF, Chowdhury ME. ED-Net: A hybrid deep learning framework for precise polyp segmentation integrating EfficientNetB0 with Double U-Net architecture and Atrous Spatial Pyramid Pooling with test-time augmentation. *Biomedical Signal Processing and Control.* 2026 Sep 1; 123:110566. [***DOI***]

Conference Proceedings

[1] Sakib SN, Jyoti O, **Kundu RK**. Advancing Cybersecurity: A Study on Phishing Website Detection Using Machine Learning. In *2025 28th International Conference on Computer and Information Technology (ICCIT)*, 2025, Dec 19 (pp. 3530-3535). IEEE. [***DOI***]

[2] Prince RH, Goni MO, Peyal HI, **Kundu RK**. Beyond the Leaf: Intelligent Lightweight Disease Detection in Tea Plants with Spatial Convolutional Neural Networks and Explainable AI. In *2025 IEEE 7th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, 2025, Dec 11 (pp. 1-6). IEEE. [***DOI***]

Manuscripts Under Review & Submitted

[1] **Kundu RK**, Mia MM, Khandakar A, Shi Z, Kibria HB, Ayari MA. Demographic fairness in skin lesion segmentation: Lesion-type confounding of age-related performance disparities. *IEEE Open Journal of the Computer Society.* 2026. **[Under review]**

[2] Kibria HB, **Kundu RK**, Kibria NBG. Data leakage in clinical machine learning: A systematic review of sources, impacts, and mitigation strategies, 2026. **[Submitted]**

[3] **Kundu RK**, Mia MM, Ayari MA, Reaz MBI, Refaat SS, Rohouma W, Khandakar A. Behavior-specific hierarchical sensor-vision fusion for cow-independent dairy cattle behavior recognition. 2026. **[Submitted]**

Research Experience

Research Collaboration (Remote) **June 2026 – Oct 2026**
Collaborator: *Dr. Amith Khandakar, Department of Electrical Engineering* *Qatar University*

Cow-Independent Multimodal Cattle Behavior Recognition ***Submitted***

- Developed **RG-ROCKET**, a MiniROCKET-based generalist with uncertainty-weighted rhythm guidance, and integrated it into a hierarchical sensor-vision framework for cow-independent cattle behavior recognition under severe class imbalance and incomplete visual observations.
- Evaluated generalization to unseen cows under strict leave-one-cow-out validation and stress-tested robustness to missing camera views, localization error, and visual perturbations.

Demographic Fairness in Skin Lesion Segmentation ***Under Review***

- Investigated whether an apparent age-related disparity in skin lesion segmentation persisted after accounting for differences in clinical case mix, using age-targeted interventions and a confound-aware statistical audit.
- Found that lesion-type composition largely explained the observed disparity, reducing the age-associated effect by **83%** and rendering it non-significant, with validation across multiple segmentation architectures and human inter-annotator agreement.

Undergraduate Thesis Research **July 2024 – June 2025**
***Supervisor**: Hafsa Binte Kibria, Department of ECE* *RUET, Bangladesh*

ED-Net: Colorectal Polyp Segmentation ***Published***

- Developed **ED-Net**, combining a pretrained EfficientNetB0 encoder with a Double U-Net architecture, ASPP for multi-scale context, SE attention, and test-time augmentation for colorectal polyp segmentation, emphasizing cross-dataset generalization, segmentation accuracy, and computational efficiency.
- Evaluated the model on Kvasir-SEG and three unseen external datasets without retraining, and analyzed model behavior using Grad-CAM and SE-block attention.

Independent & Collaborative Research **2025 – 2026**

Data Leakage in Clinical Machine Learning ***Submitted***

- Conducted a PRISMA-guided systematic review of 50 studies to examine how data leakage arises across target construction, preprocessing, validation, and patient-level partitioning in clinical machine learning.
- Combined the review with controlled tabular-data experiments to assess performance inflation under different leakage scenarios and derive stage-specific mitigation strategies to prevent leakage.

Relevant Coursework

- **CS and AI:** Data Structures & Algorithms; Object-Oriented Programming; Database Systems; Operating Systems; Computer Networks; Digital Image Processing; Neural Networks & Fuzzy Systems.
- **Mathematics and Signal Processing:** Linear Algebra; Statistics; Numerical Methods; Digital Signal Processing; Digital Techniques.

Standardized Test Scores

IELTS Academic: Overall **7.0**; Speaking 7.0; Listening 7.5; Reading 7.5; Writing 6.0 *8 November 2025*

Technical Skills

Programming:	Python, MATLAB, C, C++, Java.
ML & DL:	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV.
Data:	NumPy, pandas, Matplotlib, Jupyter.
Explainability:	SHAP, Grad-CAM.
Research Tools:	Google Colab, Kaggle, Git, GitHub, LaTeX, Mendeley.

Projects

1. **RUET-ASSR Official Website:** Led a team to build the official website for the Astronomy & Science Society of RUET using *Next.js*, *Prisma*, and *MongoDB*. [Link]
2. **Home Automation System:** Designed an IoT-based home automation system using ESP32 with remote control via Wi-Fi, developed during ECE-3212: Peripherals & Interfacing Sessional. [Link]

Awards and Achievements

1. **Technical Scholarship**, ***University Grants Commission (UGC), Bangladesh***: awarded for three consecutive years (2022, 2023, and 2024) based on outstanding B.Sc. academic performance.
2. **General Grade Scholarships**, ***Jashore Education Board, Bangladesh:*** awarded in both the SSC (Grade 10) and JSC (Grade 8) examinations for outstanding academic performance.
3. **Winner, Photography Competition**, IEEE PES Day 2024, IEEE RUET Student Branch. [Certificate]
4. **1st Runner-up, Inter-Department Sports Carnival (Badminton)**, RUET, 2025.[Certificate]

Academic Leadership & Service

1. **Chair**, IEEE RUET Signal Processing Society (SPS) Student Branch Chapter (2024–2025) [Link]
2. **Vice President (Information)**, Astronomy & Science Society of RUET (ASSR) (2024–2025) [Link]
3. **Secretary, Website & CMT Management**, IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON), 2024 (2023–2024) [Link]
4. **Webmaster**, IEEE RUET Student Branch (2023–2024) [Link]
5. **Team Leader**, AURORA National Science Festival 2023 (ASSR) [Link]

References

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